中文文档

巴适声工厂・隐私版 (Bashi Voice Factory Privacy Edition)

License MIT | version 0.1.0 | python 3.12 embed | platform Windows

版本： 0.1.0

完全离线运行的本地语音工厂网页应用。首次启动联网下载完依赖与模型之后，所有文字转语音、语音转文字、音频生成、文件保存都在你自己的电脑上完成，没有任何音频数据上传云端。

> ⚡ 想要云端语音质量、更快的合成速度？请关注 巴适声工厂 · 极速版 (Bashi Voice Factory Turbo) — 微软 Edge TTS 引擎、14 种语言、5 万字长文。需要联网。

作者： Alex Li (ncorecpu@gmail.com) 许可： MIT License

✨ 核心亮点

🔒 完全本地文字转语音

•Qwen3-TTS-12Hz-1.7B-CustomVoice 本地推理 — 合成过程零云端调用。
•9 个精选音色，内置母语试听样例。
•10 种语言：中文（普通话、粤语、四川话、北京话、东北话）、英文（美/英）、日语、韩语、德语、法语、俄语、葡萄牙语、西班牙语、意大利语。
•自然语言指令控制 — 风格、口音、情绪都可通过 instruct 提示调整。

🎯 后端自动选择

•GGUF + Vulkan：AMD / Intel / 集成显卡用户默认路线（速度快、内存友好）。
•PyTorch + CUDA：NVIDIA 用户进阶路线（权重需单独下载）。
•CPU 回退：入门级硬件（如 Intel N100 系列）也能跑。
•一键 测速 校准 ETA 估算，让等待时间贴合你这台机器。

🎙️ 本地离线音视频转文字 (STT)

•SenseVoice Small（默认，242 MB）— 中英日韩粤多语种，非自回归架构，CPU 上速度快。
•Parakeet TDT 0.6B（可选，661 MB）— 英文专用，NVIDIA 出品，WER 约 1.7%。
•Silero VAD 精准分段 — 时间戳准确、零重叠、零卡顿。
•实时滚屏字幕 SSE 流式输出；支持导出 TXT、SRT、VTT。

🛡️ 智能首次下载，断点续传

•HTTP Range / resume 内置 — 一次 WiFi 闪断不会让 2.2 GB 的 GGUF 下载从零开始。

•3 次重试，指数退避 — pip 依赖与模型文件都有同样的兜底逻辑。
•国内友好镜像：自动识别中国时区使用阿里云 PyPI；GGUF 运行模型走 ModelScope；STT 模型走 hf-mirror.cn。

📱 局域网共享，手机平板可访问

•首次启动询问是否绑定 `0.0.0.0`，允许同一 WiFi 下其他设备访问。

•手机浏览器输入电脑的局域网 IP 即可使用，跟本地网页一样流畅。

🚀 快速上手

最简流程

1.把下载的 zip 解压到任意文件夹（桌面或随便哪里都行）。

2.在解压后的顶层目录里 双击 `Start_启动.bat`。

3.首次启动会：

•安装 Python 依赖（约 700 MB，根据网速 / CPU 不同需要 8-15 分钟）

•询问是否下载 GGUF 模型（约 2.2 GB，2-15 分钟）

•自动在默认浏览器打开 `http://127.0.0.1:5050`
4.首次完成后，之后每次启动都是离线，5 秒左右即可打开。

进阶

如果想跳过顶层启动器，可以直接运行 `bashi-privacy-app\run_portable.bat`，效果完全一样。

🌐 网络行为说明

隐私承诺：本程序不主动联网、不上传音频、不收集使用数据。

仅首次启动（一次性，约 700 MB + 约 2.2 GB）：

•pip 依赖来自 `https://mirrors.aliyun.com/pypi/simple/`（海外用户默认 PyPI）
•GGUF 模型来自 `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime`
•STT 模型（用户主动点击下载）来自 `https://hf-mirror.com/csukuangfj/...`（备用 HuggingFace）

运行时网络行为：零。文字转语音、音视频转文字、音频导出全部本机完成。

唯一可选联网：UI 右下角"查看最新版本"按钮，由你主动点击时打开 files.fm 发布页。程序不自动检查更新，不在后台联网。

💻 硬件要求

下表是作者实测的数据。首次运行后点右侧面板的"测速"按钮，程序会根据你这台机器实际表现重新校准 ETA 估算 — 不用照搬此表。

硬件	自动选择的后端	"你好。" 探测（25 字）	1,000 字合成预计	状态
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	约 3 秒	几分钟	✓ 2026-05 实测
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 秒	约 5-10 分钟	√ 实测
Intel N305 笔记本 + UHD 集显	GGUF + Vulkan / DirectML	53 秒	25-46 分钟	✓ 2026-05-25 实测
Intel N100 迷你主机 + UHD 集显	GGUF + Vulkan / DirectML	126 秒	58 分钟 - 1 小时 49 分	✓ 2026-05-25 实测
NVIDIA RTX / Apple Silicon / Intel Arc	—	暂未实测	暂未实测	v0.1 未验证

> ⚠ 入门级 CPU（Intel N100 / N305 一类）仅适合短句试用 — 单次生成建议不超过 200 字。 这类机器跑 5,000 字长文需要 2-9 小时。如果想做长文音频（讲座、有声书），请用独立显卡硬件（AMD RX 500/600/9000 系、NVIDIA RTX 类）。
> ℹ Windows 上的 NVIDIA 用户也可以走 GGUF + Vulkan 路线（NVIDIA 驱动支持 Vulkan）。PyTorch + CUDA 路线在 v0.1 需要手动配置权重，不会自动启用。

🐛 常见问题排查

现象	可能原因 / 处理
双击 `Start_启动.bat` 报 "Array index expression is missing"	旧版 zip。请从 files.fm 重新下载最新版；该 BOM 问题在 v0.1.0 正式版已修复
pip 安装途中 WiFi 闪断后中止	自动重试 3 次（5/30/120 秒退避）。3 次都失败时，修好网络后重新运行启动器即可（pip 会跳过已装好的包）
GGUF 下载中断	同样自动重试，且支持 HTTP Range 续传，重新运行启动器从 `.part` 文件续传
启动报 "No usable backend was found"	查 `launch_log.txt`。常见原因：GGUF 运行 DLL 缺失、显卡驱动过旧、可用内存 <8 GB。程序会打印中英双语提示告知具体原因
STT 下载闪过 "镜像失败"	v0.1.0 正式版不会出现 — 是旧版残留的 ModelScope 路径，已删除

完整日志路径： `bashi-privacy-app\launch_log.txt`

📦 zip 包目录结构

```
bashi-voice-factory-privacy-v0.1.0/
├── Start_启动.bat                                    ← 双击这里
├── README.md                                         ← 英文文档
├── README_CN.md                                      ← 本文件
├── LICENSE                                           ← MIT 许可证
├── VERSION                                           ← 发布版本号
├── 巴适声工厂隐私版使用手册_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf
│                                                   ← 中英双语帮助 PDF
├── bashi-privacy-app/                                ← 程序代码 + 嵌入式 Python
│   ├── run_portable.bat                              ← 等效启动器（直接路径）
│   ├── README.md                                     ← 项目说明副本
│   └── ...
└── vulkan_backend_spike/                             ← GGUF 运行时（首次启动自动填
充）
    └── Qwen3-TTS-GGUF/
```

📄 许可

第三方组件保留各自原始许可：

•Qwen3-TTS-12Hz-1.7B-CustomVoice：通义实验室（阿里巴巴）Apache 2.0
•GGUF runtime：基于 HaujetZhao/Qwen3-TTS-GGUF
•SenseVoice Small / Silero VAD / Parakeet TDT：详见各自模型卡

👤 作者

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欢迎通过 GitHub release 页面或邮件提交 issue、反馈或功能建议。

English

Bashi Voice Factory Privacy Edition (巴适声工厂・隐私版)

License MIT | version 0.1.0 | python 3.12 embed | platform Windows

Version: 0.1.0

A fully offline desktop web app for high-quality text-to-speech and speech-to-text. After the one-time first-launch download, everything runs on your own machine — TTS synthesis, audio export, transcription, and storage. No audio data ever leaves your computer.

> ⚡ Need cloud-quality voices and faster generation? Check out Bashi Voice Factory Turbo (巴适声工厂・极速版) — Microsoft Edge TTS, 14 languages, 50,000-character long text. Requires internet.

Author: Alex Li (ncorecpu@gmail.com) License: MIT License

✨ Highlight Features

🔒 Fully Local Text-to-Speech

•Local Qwen3-TTS-12Hz-1.7B-CustomVoice running entirely on your machine — no cloud calls during synthesis.
•9 curated voice presets with native-language sample previews built in.
•10 languages supported: Chinese (Mandarin, Cantonese, Sichuanese, Beijing, Northeast), English (US/UK), Japanese, Korean, German, French, Russian, Portuguese, Spanish, Italian.
•Instruction control for style, accent, emotion via natural-language prompts.

🎯 Automatic Backend Selection

•GGUF + Vulkan for AMD / Intel / iGPU users (default, fast, RAM-friendly).
•PyTorch + CUDA for NVIDIA users (advanced setup; weights download separately).
•CPU fallback for entry-level hardware (works on Intel N100-class boxes).
•One-click speed test calibrates ETA estimates to your specific machine.

🎙️ Local Offline Speech-to-Text (STT)

•SenseVoice Small (default, 242 MB) — multilingual zh/en/ja/ko/yue, non-autoregressive (fast on CPU).
•Parakeet TDT 0.6B (optional, 661 MB) — English specialist, NVIDIA's ~1.7% WER champion.
•VAD-based segmentation via Silero — accurate timestamps, no chunk-boundary stutter.
•Live transcript streaming via SSE; exports to TXT / SRT / VTT.

🛡️ Smart First-Launch Download with Resume

•HTTP Range / resume built in — a transient WiFi drop won't restart your 2.2 GB GGUF download from zero.

•3-attempt retry with exponential backoff for both pip dependencies and model files.
•China-friendly mirrors: Aliyun PyPI mirror auto-detected by locale; GGUF runtime hosted on ModelScope; STT models hosted on hf-mirror.cn.

📱 LAN Sharing for Phone / Tablet Use

•First-launch prompt: bind to `0.0.0.0` to allow access from any device on the same WiFi.

•Compatible with mobile browsers — type the LAN IP and use it like a local web app.

🚀 Quick Start

Easiest path

1.Unzip the downloaded file to any folder (Desktop or wherever you like).

2.Double-click `Start_启动.bat` at the top level of the extracted folder.

3.First launch will:

•Install Python dependencies (~700 MB, 8-15 min depending on network/CPU)

•Prompt to download the GGUF model (~2.2 GB, 2-15 min depending on network)

•Open `http://127.0.0.1:5050` in your default browser
4.After first launch, all subsequent launches are offline and start in ~5 seconds.

Advanced

Users who prefer to bypass the top-level launcher can run `bashi-privacy-app\run_portable.bat` directly. Both paths are identical.

🌐 Network Behavior

Privacy commitment: this app does not phone home, does not upload audio, does not collect telemetry.

First launch only (one-time, ~700 MB + ~2.2 GB):

•pip dependencies from `https://mirrors.aliyun.com/pypi/simple/` (or PyPI default outside China)
•GGUF model from `https://modelscope.cn/models/gtree592/bashi-qwen3-tts-1.7b-customvoice-gguf-runtime`
•STT model (when user opts in) from `https://hf-mirror.com/csukuangfj/...` (or HuggingFace fallback)

Runtime network activity: zero. TTS synthesis, STT transcription, and audio export are 100% on-device.

Optional, user-initiated only: the UI footer "Check for updates" button opens a files.fm release page in a new tab. No automatic version polling; no background HTTP traffic.

💻 Hardware Expectations

The table below shows actually measured numbers from author hardware. The built-in speed test (click 测速 / Speed Test on the right-hand panel) calibrates the ETA panel to your specific machine on first use, so you don't have to extrapolate from this table.

Hardware	Auto-selected backend	"你好。" probe (25 chars)	1,000-char synthesis ETA	Status
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	~3 s	a few minutes	✓ Tested 2026-05
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 s	~5-10 min	✓ Tested
Intel N305 laptop + UHD iGPU	GGUF + Vulkan / DirectML	53 s	25-46 min	✓ Tested 2026-05-25
Intel N100 mini-PC + UHD iGPU	GGUF + Vulkan / DirectML	126 s	58 min - 1h49 min	✓ Tested 2026-05-25
NVIDIA RTX / Apple Silicon / Intel Arc	—	not yet measured	not yet measured	not validated in v0.1

> ⚠ Entry-level CPUs (Intel N100 / N305 class) are only practical for short text — under ~200 characters per generation. A 5,000-character essay would take 2-9 hours on these boxes. For long-form audio (lectures, audiobooks), please use discrete-GPU hardware (AMD RX 500/600/9000 series, NVIDIA RTX class).
> ℹ NVIDIA users on Windows can also use the GGUF + Vulkan path (NVIDIA cards support Vulkan via the proprietary driver). The PyTorch + CUDA path in v0.1 requires manual weight setup; it is not auto-configured.

🐛 Troubleshooting

Symptom	Likely cause / fix
`Start_启动.bat` shows "Array index expression is missing"	Old zip — re-download the latest from files.fm; the BOM bug was fixed in v0.1.0 final
pip install stops mid-way after WiFi blip	Retry triggers automatically (3 attempts, 5s/30s/120s backoff). If all 3 fail, fix network and re-run the launcher — pip caches what's already installed
GGUF download	Same: retries with HTTP Range/resume; re-run launcher to continue

Symptom	Likely cause / fix
interrupted	from `.part` file
App exits with "No usable backend was found"	Check `launch_log.txt` — usually GGUF runtime DLL missing, GPU driver outdated, or RAM <8 GB. App now prints a bilingual advisory with specific causes
STT download says "镜像失败"	Should not happen in v0.1.0 final — old behavior from removed ModelScope path

Full log path: `bashi-privacy-app\launch_log.txt`

📦 What's in the Zip

```
bashi-voice-factory-privacy-v0.1.0/
├── Start_启动.bat                                     ← double-click here
├── README.md                                           ← this file
├── README_CN.md                                        ← Chinese version
├── LICENSE                                             ← MIT license
├── VERSION                                             ← release version
├─── 巴适声工厂隐私版使用手册_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf
│                                                                 ← help PDF (bilingual)
├── bashi-privacy-app/                                  ← app code + embedded
Python
│   ├── run_portable.bat                                ← same launcher, direct
path
│   ├── README.md                                       ← project README copy
│   └── ...
└── vulkan_backend_spike/                               ← GGUF runtime (populated
on first launch)
    └── Qwen3-TTS-GGUF/
```

📄 License

Bundled third-party components retain their original licenses:

- Qwen3-TTS-12Hz-1.7B-CustomVoice: Tongyi Lab (Alibaba), Apache 2.0
- GGUF runtime: based on HaujetZhao/Qwen3-TTS-GGUF
- SenseVoice Small / Silero VAD / Parakeet TDT: see respective model cards

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Issues, feedback, and feature requests welcome via the GitHub release page or email.